IVB Decomposition for the Synthetic Difference-in-Differences Estimator

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1 Introduction

This note derives the Included Variable Bias (IVB) decomposition for the Synthetic Difference-in-Differences (SDiD) estimator of Arkhangelsky et al. (2021). When a researcher includes a time-varying covariate Z that may be both a confounder and post-treatment, the estimate changes. We decompose this change into two components:

$$\text{IVB} = \underbrace{\text{CE}}_{\text{Coefficient Effect}} + \underbrace{\text{RE}}_{\text{Reweighting Effect}}$$

The Coefficient Effect (CE) captures bias from conditioning on Z with fixed weights—the direct analogue of the TWFE formula $\text{IVB} = -\theta^\star \times \pi$. The Reweighting Effect (RE) captures the additional change from Z distorting the synthetic weights. RE is new and specific to adaptive-weight estimators like SDiD.

This note establishes the mathematical foundation. Conditions for RE = 0 (Step 2), practical diagnostics (Step 3), and applied illustration (Step 4) are deferred to subsequent notes.

2 Setup and Notation

2.1 Panel data

Consider a balanced panel with N units observed over T periods. Let N_0 denote control units and $N_1 = N - N_0$ treated units. Let T_0 denote pre-treatment periods and $T_1 = T - T_0$ post-treatment periods. Define:

- Y_{it}: outcome for unit i at time t
- D_{it}: treatment indicator; $D_{it} = 1$ iff $i > N_0$ and $t > T_0$ (block treatment assignment)
- Z_{it}: a scalar time-varying covariate

Collect outcomes in the $N \times T$ matrix $\mathbf{Y}$ with entries Y_{it}, and similarly $\mathbf{Z}$ for the covariate.

2.2 SDiD weights

The SDiD estimator uses two weight vectors:

- **Unit weights**: $\omega = (\omega_1, \ldots, \omega_{N_0})' \in \Delta_{N_0}$, where Δ_{N_0} is the unit simplex
- **Time weights**: $\lambda = (\lambda_1, \ldots, \lambda_{T_0})' \in \Delta_{T_0}$

These weights are chosen to minimize a regularized objective that balances pre-treatment trends between a weighted combination of controls and the treated units (Arkhangelsky et al. 2021, Algorithm 1).

Define the **extended weight vectors** that span all units and periods:

$$\tilde{\omega} = (-\omega_1, \ldots, -\omega_{N_0}, \underbrace{1/N_1, \ldots, 1/N_1}_{N_1})' \in \mathbb{R}^N \tag{1}$$

$$\tilde{\lambda} = (-\lambda_1, \ldots, -\lambda_{T_0}, \underbrace{1/T_1, \ldots, 1/T_1}_{T_1})' \in \mathbb{R}^T \tag{2}$$

Note that treated units receive uniform weight $1/N_1$ and post-treatment periods receive uniform weight $1/T_1$, following the SDiD specification.

2.3 The collapsed form

The `synthdid` package operates on a **collapsed form** $\mathbf{Y}^c$ of dimension $(N_0 + 1) \times (T_0 + 1)$:

$$Y^c_{it} = \begin{cases} Y_{it} & \text{if } i \leq N_0 \text{ and } t \leq T_0 \\ \frac{1}{T_1} \sum_{s=T_0+1}^{T} Y_{is} & \text{if } i \leq N_0 \text{ and } t = T_0 + 1 \\ \frac{1}{N_1} \sum_{j=N_0+1}^{N} Y_{jt} & \text{if } i = N_0 + 1 \text{ and } t \leq T_0 \\ \frac{1}{N_1 T_1} \sum_{j=N_0+1}^{N} \sum_{s=T_0+1}^{T} Y_{js} & \text{if } i = N_0 + 1 \text{ and } t = T_0 + 1 \end{cases}$$

The collapsed form averages over treated units and post-treatment periods, exploiting the fact that SDiD assigns uniform weights to these dimensions. The same collapsing applies to $\mathbf{Z}$, yielding $\mathbf{Z}^c$.

Remark 1 (Collapsing preserves the bilinear form)**.** *For any extended weight vectors $\tilde{\omega}$, $\tilde{\lambda}$ and matrix $\mathbf{M}$ (either $\mathbf{Y}$ or $\mathbf{Z}$):*

$$\tilde{\omega}' \mathbf{M} \tilde{\lambda} = \tilde{\omega}^{c\prime} \mathbf{M}^c \tilde{\lambda}^c$$

where $\tilde{\omega}^c = (-\omega_1, \ldots, -\omega_{N_0}, 1)' \in \mathbb{R}^{N_0+1}$ and $\tilde{\lambda}^c = (-\lambda_1, \ldots, -\lambda_{T_0}, 1)' \in \mathbb{R}^{T_0+1}$.

This follows because the last entry of $\tilde{\omega}^c$ is 1 (not $1/N_1$), applied to the last row of $\mathbf{M}^c$ which is already the average over treated units. Similarly for $\tilde{\lambda}^c$.

2.4 Notation bridge to the IVB paper

The IVB formula for TWFE is IVB $= -\theta^\star \times \pi$, where $\theta^\star$ is the coefficient of Z in the long TWFE model and π is the coefficient of D in the auxiliary regression $Z \sim D + \text{FE}$. The SDiD analogues are:

	TWFE	**SDiD**
$Z \to Y$ coefficient	$\theta^\star$	$\hat{\beta}_h$
$D \to Z$ effect	π	$\hat{\tau}_Z$
IVB formula	$-\theta^\star \times \pi$	$-\hat{\beta}_h \times \hat{\tau}_Z$ (CE only)

3 SDiD Estimator with Covariates

3.1 Without covariates

The SDiD estimate without Z is the weighted double-difference:

$$\hat{\tau}_{\text{short}} = \tilde{\omega}'_{\text{short}} \, \mathbf{Y} \, \tilde{\lambda}_{\text{short}} \tag{3}$$

where $(\omega_{\text{short}}, \lambda_{\text{short}})$ solve the regularized optimization in Arkhangelsky et al. (2021, Algorithm 1).

3.2 With covariates

When a covariate Z is included, the SDiD estimator jointly optimizes weights and the covariate coefficient:

$$\hat{\tau}_{\text{long}} = \tilde{\omega}'_{\text{long}} \left(\mathbf{Y} - \mathbf{Z}\hat{\beta}_{\text{long}}\right) \tilde{\lambda}_{\text{long}} \tag{4}$$

where $(\omega_{\text{long}}, \lambda_{\text{long}}, \hat{\beta}_{\text{long}})$ are jointly estimated via alternating optimization: for fixed $\hat{\beta}$, update (ω, λ) by Frank-Wolfe on the residualized outcome $\mathbf{Y} - \mathbf{Z}\hat{\beta}$; for fixed (ω, λ), update $\hat{\beta}$ by gradient descent.

A key property of this optimization is that $\hat{\beta}$ is estimated from variation where $D = 0$. On the collapsed form, the optimization uses: (i) control units in pre-treatment periods ($D = 0$), (ii) control units' post-treatment average ($D = 0$), and (iii) the treated units' pre-treatment values ($D = 0$). The treated post-treatment cell enters the objective only through $\hat{\tau}$, not through $\hat{\beta}$. Therefore, $\hat{\beta}$ captures the $Z \to Y$ relationship uncontaminated by treatment.

4 Three Estimators and the IVB Decomposition

Definition 1 (Three estimators)**.** *Given a panel* $(\mathbf{Y}, \mathbf{D}, \mathbf{Z})$*:*

1. ***Short model****:* $\hat{\tau}_{short}$ *is the SDiD estimate without* Z*, with optimized weights* $(\omega_{short}, \lambda_{short})$*.*
2. ***Long model****:* $\hat{\tau}_{long}$ *is the SDiD estimate with* Z*, with jointly optimized* $(\omega_{long}, \lambda_{long}, \hat{\beta}_{long})$*.*
3. ***Hybrid model****:* $\hat{\tau}_{hybrid}$ *is the SDiD estimate with* Z *included, but with weights* fixed *at* $(\omega_{short}, \lambda_{short})$ *from the short model. Only* $\hat{\beta}_h$ *is optimized:*

$$\hat{\tau}_{hybrid} = \tilde{\omega}'_{short} \left(\mathbf{Y} - \mathbf{Z}\hat{\beta}_h\right) \tilde{\lambda}_{short} \tag{5}$$

The hybrid estimator is the bridge between the short and long models. It isolates the effect of adding Z to the regression (via $\hat{\beta}_h$) from the effect of Z on weight optimization.

Definition 2 (IVB decomposition for SDiD)**.** *The Included Variable Bias is:*

$$IVB \equiv \hat{\tau}_{long} - \hat{\tau}_{short}$$

Decompose by adding and subtracting $\hat{\tau}_{hybrid}$*:*

$$\boxed{IVB = \underbrace{\left(\hat{\tau}_{hybrid} - \hat{\tau}_{short}\right)}_{CE:\ Coefficient\ Effect} + \underbrace{\left(\hat{\tau}_{long} - \hat{\tau}_{hybrid}\right)}_{RE:\ Reweighting\ Effect}} \tag{6}$$

The decomposition is exact by construction—no approximation is involved. CE measures the change from including Z while holding weights fixed. RE measures the additional change from weight re-optimization.

5 The Coefficient Effect: An FWL-like Identity

Proposition 1 (FWL-like identity for CE)**.** *Under the SDiD estimator with a scalar covariate* Z *and fixed weights* $(\omega_{short}, \lambda_{short})$*:*

$$CE = -\hat{\beta}_h \times \hat{\tau}_Z \tag{7}$$

where:

- $\hat{\beta}_h$ *is the coefficient of* Z *in the hybrid model (Definition 1), estimated from* $D = 0$ *variation on the collapsed form;*

- $\hat{\tau}_Z \equiv \tilde{\omega}'_{short}\mathbf{Z}\,\tilde{\lambda}_{short}$ *is the SDiD double-difference applied to* Z*—i.e., the "SDiD treatment effect on* Z*," computed with the short-model weights.*

Proof. From equations (3) and (5), both $\hat{\tau}_{\text{short}}$ and $\hat{\tau}_{\text{hybrid}}$ use the same extended weight vectors $\tilde{\omega}_{\text{short}}$ and $\tilde{\lambda}_{\text{short}}$. Therefore:

$$\begin{aligned}
\text{CE} &= \hat{\tau}_{\text{hybrid}} - \hat{\tau}_{\text{short}} \\
&= \tilde{\omega}'_{\text{short}}\,(\mathbf{Y} - \mathbf{Z}\hat{\beta}_h)\,\tilde{\lambda}_{\text{short}} - \tilde{\omega}'_{\text{short}}\,\mathbf{Y}\,\tilde{\lambda}_{\text{short}} \\
&= \tilde{\omega}'_{\text{short}}\,\mathbf{Y}\,\tilde{\lambda}_{\text{short}} - \hat{\beta}_h \cdot \tilde{\omega}'_{\text{short}}\,\mathbf{Z}\,\tilde{\lambda}_{\text{short}} - \tilde{\omega}'_{\text{short}}\,\mathbf{Y}\,\tilde{\lambda}_{\text{short}} \\
&= -\hat{\beta}_h \cdot \tilde{\omega}'_{\text{short}}\,\mathbf{Z}\,\tilde{\lambda}_{\text{short}} \\
&= -\hat{\beta}_h \times \hat{\tau}_Z
\end{aligned} \tag{8}$$

The key step uses the bilinearity of the double-difference operator: $\tilde{\omega}'(\mathbf{Y} - \mathbf{Z}\beta)\tilde{\lambda} = \tilde{\omega}'\mathbf{Y}\tilde{\lambda} - \beta \cdot \tilde{\omega}'\mathbf{Z}\tilde{\lambda}$, which holds because $\hat{\beta}_h$ is a scalar (single covariate Z) and the weights $(\tilde{\omega}_{\text{short}}, \tilde{\lambda}_{\text{short}})$ are the same in both terms. □

Remark 2 (The identity is purely algebraic)**.** *The identity CE* $= -\beta \cdot \hat{\tau}_Z$ *holds for* any *scalar* β*, not just the optimized* $\hat{\beta}_h$*. The proof depends only on the bilinearity of the double-difference operator and the fact that the same weights appear in both* $\hat{\tau}_{hybrid}$ *and* $\hat{\tau}_{short}$*. The SDiD optimization procedure determines the* value *of* $\hat{\beta}_h$ *(and hence the numerical value of CE), but the product structure is a tautology of the estimator's algebraic form—analogous to how the Frisch-Waugh-Lovell identity holds regardless of the data-generating process.*

Remark 3 (Generalization to multiple covariates)**.** *With* K *covariates* $\mathbf{Z}_1, \ldots, \mathbf{Z}_K$ *and coefficient vector* $\hat{\boldsymbol{\beta}}_h = (\hat{\beta}_{h,1}, \ldots, \hat{\beta}_{h,K})'$*, the identity generalizes to:*

$$CE = -\sum_{k=1}^{K} \hat{\beta}_{h,k} \times \hat{\tau}_{Z_k}$$

where $\hat{\tau}_{Z_k} = \tilde{\omega}'_{short}\mathbf{Z}_k\tilde{\lambda}_{short}$ *is the SDiD double-difference of the* k*-th covariate. The proof follows identically from the linearity of the contract*$_3$ *operator* $\mathbf{Y} - \sum_k \mathbf{Z}_k\hat{\beta}_{h,k}$*.*

5.1 Interpretation of the two factors

$\hat{\beta}_h$ **(the** $Z \to Y$ **channel):** This coefficient captures how much Z predicts Y, conditional on unit and time structure, using only $D = 0$ variation. It reflects the "clean'' association between Z and Y uncontaminated by treatment. It is the SDiD analogue of $\theta^\star$ in the TWFE IVB formula.

$\hat{\tau}_Z$ **(the** $D \to Z$ **channel):** This is the SDiD double-difference applied to Z as if it were the outcome. It measures how much treatment changes Z, net of the SDiD-weighted parallel trends adjustment. It is the SDiD analogue of π in the TWFE IVB formula.

Sign of CE: The direction of the Coefficient Effect depends on the product $\hat{\beta}_h \times \hat{\tau}_Z$:

$\hat{\beta}_h$	$\hat{\tau}_Z$	CE	Including $Z\ldots$
$+$	$+$	$-$	reduces $\hat{\tau}$
$+$	$-$	$+$	increases $\hat{\tau}$
$-$	$+$	$+$	increases $\hat{\tau}$
$-$	$-$	$-$	reduces $\hat{\tau}$

6 Fixed Weights and the Relationship to TWFE

Corollary 1 (Fixed weights imply RE = 0)**.** *If the weight vectors (ω, λ) are pre-specified and not re-optimized when Z is included, then the Reweighting Effect is zero:*

$$RE = 0 \qquad \Longrightarrow \qquad IVB = CE = -\hat{\beta}_h \times \hat{\tau}_Z$$

Proof. When the weights are fixed, the long model uses the same (ω, λ) as the short model. By definition, the hybrid estimator also uses these weights. Therefore $\hat{\tau}_{\text{long}} = \hat{\tau}_{\text{hybrid}}$ and RE = 0. □

Remark 4 (Relationship to TWFE)**.** *In TWFE (OLS with unit and time fixed effects), there are no adaptive weights—the implicit "weights" on observations are determined by the OLS projection and the treatment structure, not by an optimization over (ω, λ). The standard IVB formula for TWFE,*

$$IVB_{TWFE} = -\theta^\star \times \pi,$$

follows directly from the Frisch-Waugh-Lovell theorem, where $\theta^\star$ is the coefficient of Z in the long TWFE model and π is the coefficient of D in the auxiliary regression $Z \sim D + \alpha_i + \mu_t$.

The SDiD decomposition generalizes *this result: in any estimator with adaptive weights, the total sensitivity to covariate inclusion has an additional component (RE) beyond the FWL-like coefficient channel (CE). TWFE is the limiting case where RE vanishes.*

Note, however, that SDiD with uniform weights is not identical *to TWFE when covariates are present. SDiD estimates $\hat{\beta}$ from the collapsed form via gradient descent, while TWFE estimates $\theta^\star$ via full-panel OLS—these are different optimization problems yielding different coefficient values (confirmed numerically in Step 0: $\hat{\beta}_h = 0.285$ vs. $\hat{\theta}_{WLS} = 0.557$). The structural parallel is in the* form *of the decomposition ($-$coefficient $\times$ effect on Z), not in the numerical values.*

The novelty of the SDiD decomposition is the RE term, which arises only in estimators with adaptive weights and has no TWFE analogue.

7 Characterization of the Reweighting Effect

7.1 Definition

The Reweighting Effect is:

$$\text{RE} = \hat{\tau}_{\text{long}} - \hat{\tau}_{\text{hybrid}} = \tilde{\omega}'_{\text{long}}(\mathbf{Y} - \mathbf{Z}\hat{\beta}_{\text{long}})\tilde{\lambda}_{\text{long}} - \tilde{\omega}'_{\text{short}}(\mathbf{Y} - \mathbf{Z}\hat{\beta}_h)\tilde{\lambda}_{\text{short}} \tag{9}$$

This expression involves three simultaneous changes between the hybrid and long models:

1. ω_{long} vs. ω_{short} (unit weight distortion)
2. λ_{long} vs. λ_{short} (time weight distortion)
3. $\hat{\beta}_{\text{long}}$ vs. $\hat{\beta}_h$ (coefficient change from re-optimization)

7.2 No closed-form expression

Unlike CE, the Reweighting Effect does not simplify to a product of two interpretable quantities. One could attempt a sub-decomposition by changing ω, λ, and $\hat{\beta}$ one at a time, but the result would depend on the order of changes (a path-dependence problem analogous to Shapley value decomposition). Since the three changes arise jointly from a single re-optimization, no natural ordering exists.

Despite the absence of a closed form, RE is always **computable**: three calls to `synthdid_estimate` (short, long, hybrid) suffice, with no latent quantities required.

7.3 Empirical pattern from Step 0

In the feasibility check (Step 0 report), weight changes were asymmetric:

- Unit weights: $\max_i |\omega_{\text{long},i} - \omega_{\text{short},i}| = 0.007$ (negligible)
- Time weights: $\max_t |\lambda_{\text{long},t} - \lambda_{\text{short},t}| = 0.111$ (substantial)

This suggests that **time-weight distortion** is the primary channel for RE in this DGP. Intuitively, including Z changes which pre-treatment periods are informative for constructing the counterfactual trend, while the cross-sectional matching (unit weights) is more robust to covariate inclusion.

7.4 Sufficient conditions for RE = 0

We now derive conditions under which the SDiD weights are invariant to covariate adjustment, so that RE = 0 and IVB = CE = $-\hat{\beta}_h \times \hat{\tau}_Z$.

The SDiD weight optimization (Arkhangelsky et al. 2021, Algorithm 1) involves two sub-problems on the collapsed form $\mathbf{M}^c = (\mathbf{Y} - \mathbf{Z}\hat{\beta})^c$:

- **Unit weights** ω: find $\omega \in \Delta_{N_0}$ and intercept c_ω minimizing a regularized MSE that matches the ω-weighted control average to the treated average across T_0 pre-treatment periods.
- **Time weights** λ: find $\lambda \in \Delta_{T_0}$ and intercept c_λ minimizing a regularized MSE that matches the λ-weighted pre-treatment average to the post-treatment average across N_0 control units.

Two algebraic properties of the SDiD weights are key:

P1. Weights sum to one: $\sum_{i=1}^{N_0} \omega_i = 1$ and $\sum_{t=1}^{T_0} \lambda_t = 1$ (simplex constraint).

P2. Intercepts: Both ω and λ optimizations include a free intercept (default in `synthdid`).

Proposition 2 (Additive Z implies RE = 0)**.** *Assume the SDiD optimization uses intercepts (`omega.intercept = lambda.intercept = TRUE`). If $Z_{it} = a_i + b_t$ for some unit-specific a_i and time-specific b_t, then:*

$$\omega_{long} = \omega_{short}, \qquad \lambda_{long} = \lambda_{short}, \qquad \hat{\beta}_{long} = \hat{\beta}_h, \qquad RE = 0.$$

Proof. It suffices to show that the ω and λ optimization objectives are invariant to the covariate adjustment $\mathbf{Z}\beta$ for any scalar β, so that the weight minimizers do not depend on whether Z is included. We treat the two components a_i and b_t separately.

Part A: Time-invariant component ($Z_{it} = a_i$).

λ objective. The λ optimization minimizes (on the collapsed form, after optimizing the intercept c_λ):

$$\frac{1}{N_0} \sum_{i=1}^{N_0} \left(r_i(\lambda) - \bar{r}(\lambda)\right)^2 + \zeta_\lambda^2 \|\lambda\|^2$$

where $r_i(\lambda) = \sum_{t=1}^{T_0} \lambda_t M_{it}^c - M_{i,T_0+1}^c$ and $\bar{r}$ is the mean across controls (the intercept absorbs the mean). When we residualize by $Z_{it} = a_i$:

$$r_i^Z(\lambda) = \sum_{t=1}^{T_0} \lambda_t (M_{it}^c - a_i\beta) - (M_{i,T_0+1}^c - a_i\beta) = r_i(\lambda) - a_i\beta \left(\sum_t \lambda_t - 1\right) = r_i(\lambda)$$

by property **P1** ($\sum_t \lambda_t = 1$). The $a_i\beta$ terms cancel exactly, so $r_i^Z = r_i$ and the λ objective is unchanged.

ω objective. The ω optimization minimizes (after optimizing c_ω):

$$\frac{1}{T_0} \sum_{t=1}^{T_0} \left(s_t(\omega) - \bar{s}(\omega)\right)^2 + \zeta_\omega^2 \|\omega\|^2$$

where $s_t(\omega) = \sum_{i=1}^{N_0} \omega_i M_{it}^c - M_{N_0+1,t}^c$. When we residualize by $Z_{it} = a_i$:

$$s_t^Z(\omega) = s_t(\omega) - \beta \left(\sum_i \omega_i a_i - \bar{a}_{\text{treated}} \right)$$

The shift $-\beta(\sum_i \omega_i a_i - \bar{a}_{\text{treated}})$ is constant across t (since a_i does not depend on t). After subtracting the mean $\bar{s}^Z = \bar{s} - \beta(\cdots)$, the residuals $s_t^Z - \bar{s}^Z = s_t - \bar{s}$ are identical. The objective is unchanged.

Part B: Unit-invariant component ($Z_{it} = b_t$).

ω objective. With $Z_{it} = b_t$, the residualized values are $M_{it}^c - b_t\beta$ for all i. Since b_t is common to all units at time t:

$$s_t^Z(\omega) = \sum_i \omega_i (M_{it}^c - b_t\beta) - (M_{N_0+1,t}^c - b_t\beta) = s_t(\omega) - b_t\beta \left(\sum_i \omega_i - 1 \right) = s_t(\omega)$$

by property **P1** ($\sum_i \omega_i = 1$). The $b_t\beta$ terms cancel exactly.

λ objective. With $Z_{it} = b_t$:

$$r_i^Z(\lambda) = r_i(\lambda) - \beta \left(\sum_t \lambda_t b_t - \bar{b}_{\text{post}} \right)$$

The shift is constant across i (since b_t does not depend on i). After demeaning (intercept), $r_i^Z - \bar{r}^Z = r_i - \bar{r}$. The objective is unchanged.

Combining. Since the ω and λ objectives are invariant to both a_i and b_t components of Z, they are invariant to $Z_{it} = a_i + b_t$. Therefore $\omega^*(\beta) = \omega^*(0) = \omega_{\text{short}}$ and $\lambda^*(\beta) = \lambda^*(0) = \lambda_{\text{short}}$ for all β.

With weights fixed at $(\omega_{\text{short}}, \lambda_{\text{short}})$, the alternating optimization converges to $\hat{\beta}_{\text{long}} = \hat{\beta}_h$ (since the β sub-problem is identical). Therefore $\hat{\tau}_{\text{long}} = \hat{\tau}_{\text{hybrid}}$ and RE $= 0$. □

Remark 5 (Special cases)**.** *Proposition 2 includes two important special cases:*

- ***Time-invariant*** *Z ($Z_{it} = a_i$, $b_t = 0$): covariates that capture permanent unit characteristics (e.g., GDP level, geographic features, regime type). RE = 0.*
- ***Unit-invariant*** *Z ($Z_{it} = b_t$, $a_i = 0$): covariates that capture common shocks (e.g., global oil price, a crisis dummy). RE = 0.*

In both cases, the two cancellation mechanisms work together: the simplex constraint ($\sum \omega_i = \sum \lambda_t = 1$) eliminates one component, and the intercept absorbs the other.

Remark 6 (Approximate RE ≈ 0 for near-additive Z)**.** *In practice, most covariates are not exactly additive. Write $Z_{it} = a_i + b_t + \varepsilon_{it}$, where ε_{it} is the within-unit-and-time residual (the "interactive" component of Z). By Proposition 2, the $a_i + b_t$ component does not affect the weights. Only the ε_{it} component can induce weight distortion.*

Therefore, RE is driven by $\hat{\beta} \times \varepsilon_{it}$—the interaction between Z's predictive power for Y (through $\hat{\beta}$) and Z's within-variation (through ε_{it}). A practical diagnostic: if a two-way fixed effects regression of Z on unit and time dummies yields high R^2, then ε_{it} is small and RE is expected to be small.

Remark 7 (When is RE large?)**.** *RE is expected to be large when:*

1. *Z has substantial interactive variation (ε_{it} large relative to $a_i + b_t$), AND*
2. *Z strongly predicts Y ($|\hat{\beta}|$ large), AND*
3. *the interactive variation is heterogeneous across units or periods in a way that shifts the optimal synthetic control weights.*

A covariate with strong unit $\times$ time interactions (e.g., bilateral trade flows, unit-specific policy responses to a common shock) is most likely to generate large RE.

8 Diagnostic Procedure

This section presents the IVB diagnostic as a structured procedure the applied researcher can follow.

8.1 The researcher's problem

The researcher has estimated a treatment effect $\hat{\tau}$ using SDiD and is considering whether to include a time-varying covariate Z_{it}. The covariate may play a dual role:

- **Confounder channel** ($Z \to D$ or $Z \to Y$): including Z removes omitted variable bias.
- **Post-treatment channel** ($D \to Z \to Y$): including Z introduces overcontrol bias by removing an indirect effect.

The researcher does not know a priori which role dominates. The IVB decomposition does not resolve this ambiguity—that requires causal knowledge—but it quantifies the *consequences* of inclusion and decomposes them into interpretable components.

8.2 Protocol

Step 1. Pre-diagnostic for RE magnitude. Before running the full decomposition, assess whether Z has substantial interactive variation:

$$Z_{it} \sim \alpha_i + \mu_t + \varepsilon_{it} \qquad \text{(TWFE regression of } Z\text{)}$$

If R^2_Z is high (e.g., > 0.90), Proposition~2 implies RE will be small and IVB $\approx$ CE. The full decomposition may still be run as confirmation, but the researcher can anticipate that the FWL-like formula will dominate.

Step 2. Compute the three estimates.

(a) $\hat{\tau}_{\text{short}} \leftarrow \texttt{synthdid_estimate}(\mathbf{Y}, N_0, T_0)$

(b) $\hat{\tau}_{\text{long}} \leftarrow \texttt{synthdid_estimate}(\mathbf{Y}, N_0, T_0, X = \mathbf{Z})$

(c) Extract $\omega_{\text{short}}, \lambda_{\text{short}}$ from (a). Then:

$$\hat{\tau}_{\text{hybrid}} \leftarrow \texttt{synthdid_estimate}(\mathbf{Y}, N_0, T_0, X = \mathbf{Z}, \text{weights} = (\omega_{\text{short}}, \lambda_{\text{short}}), \text{update} = \texttt{FALSE})$$

Step 3. Decompose.

$$\begin{aligned} \text{IVB} &= \hat{\tau}_{\text{long}} - \hat{\tau}_{\text{short}} \\ \text{CE} &= \hat{\tau}_{\text{hybrid}} - \hat{\tau}_{\text{short}} \\ \text{RE} &= \hat{\tau}_{\text{long}} - \hat{\tau}_{\text{hybrid}} \end{aligned}$$

Extract $\hat{\beta}_h$ from the hybrid fit and compute $\hat{\tau}_Z = \tilde{\omega}'_{\text{short}} \mathbf{Z} \tilde{\lambda}_{\text{short}}$.

Verify: CE $= -\hat{\beta}_h \times \hat{\tau}_Z$ (numerical check; should hold to machine precision).

Step 4. Report. Present the following table:

	Value	Share of IVB
$\hat{\tau}_{\text{short}}$ (without Z)		
$\hat{\tau}_{\text{long}}$ (with Z)		
IVB $= \hat{\tau}_{\text{long}} - \hat{\tau}_{\text{short}}$		100%
CE (coefficient effect)		%
RE (reweighting effect)		%
$\hat{\beta}_h$ ($Z \to Y$ in hybrid)		
$\hat{\tau}_Z$ ($D \to Z$ via SDiD)		

8.3 Interpretation guide

The five reported quantities tell a coherent story:

1. **|IVB| small relative to SE**($\hat{\tau}$): Including or excluding Z does not materially change the estimate. Report either specification; note the robustness.
2. **|IVB| large, CE-dominated** (|CE/IVB| > 0.7): The change is driven by the coefficient channel ($-\hat{\beta}_h \times \hat{\tau}_Z$). The researcher can reason about the two factors:
   - $\hat{\beta}_h$ reflects how strongly Z predicts Y (conditional on unit/time structure). This is estimable and interpretable.
   - $\hat{\tau}_Z$ reflects how much treatment changes Z. If treatment *should* change Z (mediation), CE captures overcontrol. If Z changes for confounding reasons, CE captures bias removal.
   - The sign of $\hat{\beta}_h \times \hat{\tau}_Z$ reveals the direction (see sign table in Section 4).
3. **|IVB| large, RE-dominated** (|RE/IVB| > 0.7): The change is mainly from weight distortion—including Z changes which units/periods form the counterfactual. The researcher should:
   - Examine whether the reweighted synthetic control has better or worse pre-treatment fit.
   - Check the R^2_Z diagnostic: low R^2_Z (high interactive variation in Z) is consistent with large RE.
   - Consider whether the weight change is substantively desirable (e.g., Z helps target a more appropriate donor pool) or an artifact.
4. **CE and RE of opposite signs**: Partial cancellation. The total IVB may be small even though each component is large. This masking is invisible without the decomposition and warrants careful examination—the "right" answer depends on which channel the researcher trusts.

8.4 What the diagnostic does NOT do

The decomposition is a *sensitivity analysis*, not a specification test. It cannot:

- Determine whether Z is a confounder, a mediator, or both. That requires causal reasoning beyond the data.
- Identify the "correct" specification. Both $\hat{\tau}_{\text{short}}$ and $\hat{\tau}_{\text{long}}$ are valid under their respective identifying assumptions.
- Replace domain knowledge about the causal structure connecting D, Z, and Y.

What it *can* do is quantify the stakes: how much does the choice matter, and through which channel? This converts an opaque modeling choice into a transparent, interpretable trade-off.

8.5 Inference

Standard errors for $\hat{\tau}_{\text{short}}$ and $\hat{\tau}_{\text{long}}$ are available from the `synthdid` package via placebo variance estimation (Arkhangelsky et al. 2021, Section 5). For inference on the decomposition components:

1. **Placebo-based**: Apply the full diagnostic procedure (Steps 2–3) to each placebo draw (permuting treatment assignment among control units). The distribution of placebo CE and RE provides standard errors and confidence intervals for both components.
2. **Jackknife**: For each control unit j, recompute the three estimates dropping unit j. The jackknife variance of CE and RE captures sensitivity to individual units.

In both cases, the key question is whether |IVB| is large relative to the sampling uncertainty in $\hat{\tau}_{\text{short}}$. If $|\text{IVB}| \ll \text{SE}(\hat{\tau}_{\text{short}})$, the decomposition is informative but the specification choice is immaterial for inference.

8.6 Numerical verification (Step 0)

The feasibility check confirmed all identities on a DGP with $N = 40$, $T = 20$, Z purely post-treatment:

Quantity	Value	Check
$\hat{\tau}_{\text{short}}$	1.673	
$\hat{\tau}_{\text{long}}$	1.151	
$\hat{\tau}_{\text{hybrid}}$	1.362	
IVB	-0.522	
CE	-0.311	$= -\hat{\beta}_h \times \hat{\tau}_Z = -0.285 \times 1.089$ (error $< 10^{-15}$)
RE	-0.211	$\neq 0$ (weights changed)
CE share	59.6%	
RE share	40.4%	

9 Applied Illustration: RDD-Trade

We apply the IVB diagnostic to the author's paper on foreign policy alignment when China overtakes the United States as a country's top trade partner (currently in R&R). The treatment is Brazil ($N_1 = 1$), the treatment year is 2008 ($T_0 = 12$, $T_1 = 7$), and the outcome is UNGA voting distance from China. The panel has $N = 96$ units and $T = 19$ periods.

The problematic covariate is current account deficit as a share of GDP (CA_GDP). It may act as a confounder (the 2008 crisis affected both trade patterns and political alignment) or as post-treatment (treatment changes trade flows, which affect the deficit).

9.1 Pre-diagnostic

Following the protocol in Section 7, we first assess whether RE is expected to be large:

$$R^2(\text{CA_GDP} \sim \alpha_i + \mu_t) = 0.494$$

This moderate R^2 indicates that roughly half of the variation in CA_GDP is interactive (ε_{it}), so RE may be substantial (Proposition~2).

9.2 Results

We run three analyses: (1) CA_GDP as the only covariate, (2) all 13 covariates used in the paper, and (3) the marginal effect of adding CA_GDP to the existing specification.

	CA_GDP only (Analysis 1)	**All covariates** (Analysis 2)	**Marginal** (Analysis 3)
$\hat{\tau}_{\text{short}}$	-0.2720	-0.2720	-0.2585
$\hat{\tau}_{\text{long}}$	-0.2685	-0.2639	-0.2639
$\hat{\tau}_{\text{hybrid}}$	-0.2699	-0.2874	-0.2677
IVB	0.004	0.008	-0.005
CE	0.002 (60.7%)	-0.015 (-191%)	-0.009 (169%)
RE	0.001 (39.3%)	0.023 (291%)	0.004 (-69%)
$\hat{\beta}_h$ (CA_GDP)	0.011	-0.019	–
$\hat{\tau}_Z$ (CA_GDP)	-0.202	-0.202	–

All decomposition identities and FWL checks pass to machine precision.

9.3 Interpretation

The inclusion of CA_GDP has a negligible effect on the SDiD estimate. Across all three analyses, $|\text{IVB}| < 0.01$, compared to $\hat{\tau}_{\text{short}} \approx -0.272$ (SE ≈ 0.133 from the paper). The IVB is less than 3% of $\hat{\tau}$ and less than 8% of the standard error. The specification choice—include or exclude CA_GDP—is immaterial for inference.

Three additional findings:

1. **Analysis 2 reveals cancellation.** When all 13 covariates are included, CE and RE have opposite signs and partially cancel: CE pulls the estimate down (-0.015) while RE pushes it up ($+0.023$). The total IVB ($+0.008$) masks large offsetting forces. This cancellation is invisible without the decomposition.

2. **CA_GDP is not the dominant covariate.** In the per-covariate CE decomposition (Analysis 2), the largest contributor is government deficit (`govdef_GDP`, $\text{CE}_k = -0.016$), not CA_GDP ($\text{CE}_k = -0.004$). Trade with China (`perc_trade_with_china`, $\text{CE}_k = +0.004$) partially offsets.

3. **Weight distortion is small.** Maximum unit weight change is 0.005 and maximum time weight change is 0.048 (Analysis 2). The synthetic control is stable across specifications.

Remark 8 (Practical implication for the RDD-Trade paper)**.** *The IVB diagnostic confirms that the main SDiD result ($\hat{\tau} \approx -0.27$, $p = 0.011$) is robust to the inclusion or exclusion of current account deficit. The researcher can report either specification with confidence that the choice does not affect the substantive conclusion.*

10 References

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic Difference-in-Differences. *American Economic Review*, 111(12), 4088–4118.

Caetano, C., & Callaway, B. (2024). Difference in Differences when Parallel Trends Holds Conditionally. arXiv:2406.15288.

Lin, Z., & Zhang, Y. (2022). Covariate Effect Bias in Dynamic Two-Way Fixed Effects Models. *Economics Letters*, 221, 110902.